



MANIPAL

ACADEMY *of* HIGHER EDUCATION

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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5151 - LAB 2

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MANIPAL SCHOOL OF INFORMATION SCIENCES

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Aim:

We are using Wireshark to investigate the transmission of dummy login credentials over HTTP and HTTPS.

Lab Setup:

Attacker	Kali Linux
Victim	Ubuntu on Docker (Mac)
HTTP/HTTPS endpoints	176.16.241.1/index.php on port 80/443
Creds	Admin/badpassword
Capturing interface on Wireshark	eth0

Experiment:

Spinning up the Ubuntu Machine on docker:

```

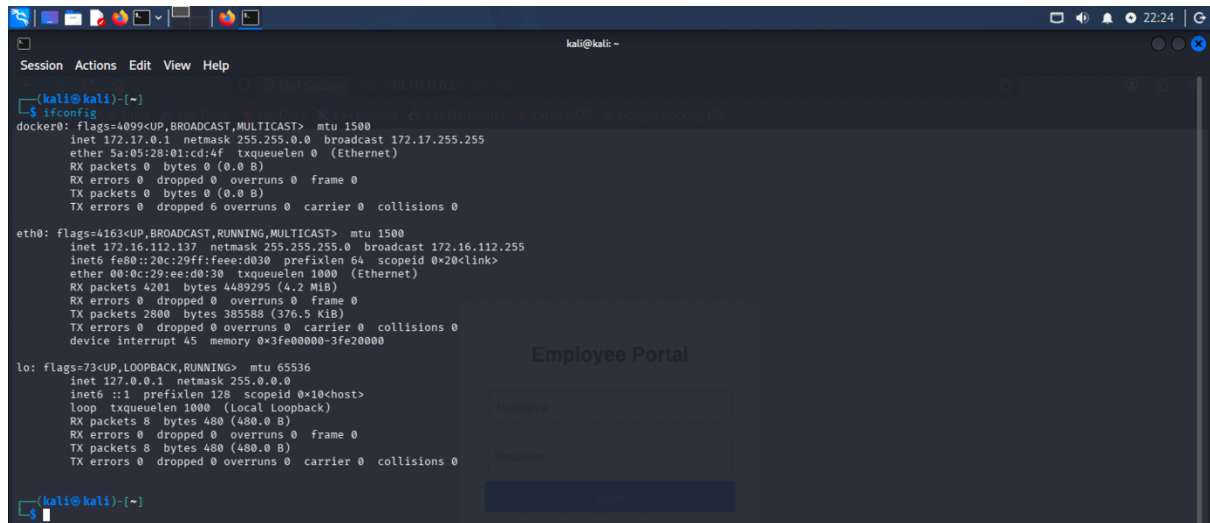
Terminal Shell Edit View Window Help
wireshark_lab -- root@9faa32f7cc3b: / -- -zsh -- 158x43

haris@M1-MacBook wireshark_lab % docker compose build
WARN[0000] /Users/haris/Developer/MSIS Manipal/SEM 1/CYT/Assignments/Lab/2_Lab/manipal_labs/labs/wireshark_lab/docker-compose.yml: the attribute 'version' is
obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 1.0s (11/11) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 7068                                           0.0s
=> [internal] load build definition from Dockerfile                     0.0s
=> => transferring dockerfile: 865B                                       0.0s
=> [internal] load metadata for docker.io/library/msis_labs:latest      0.1s
=> [internal] load dockerignore                                          0.0s
=> => transferring context: 2B                                            0.0s
=> [1/5] FROM docker.io/library/msis_labs:latest@sha256:809fa9952a89ce7de9264c462759bcdcf204d0168c9c67941980108846ea8681c
=> resolve docker.io/library/msis_labs:latest@sha256:809fa9952a89ce7de9264c462759bcdcf204d0168c9c67941980108846ea8681c
=> CACHED [2/5] RUN apt-get update && apt-get install -y apache2 php libapache2-mod-php openssl && apt-get clean
=> CACHED [3/5] RUN a2enmod ssl && a2ensite default-ssl
=> CACHED [4/5] RUN openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout /etc/ssl/private/ssl-cert-snakeoil.key -out /etc/ssl/certs/ssl
=> CACHED [5/5] RUN rm /var/www/html/index.html
=> exporting to image
=> => exporting layers                                                    0.0s
=> => exporting manifest sha256:9e82b828703dadb9f6f51c14f57f98e38636d6291362d837c1dc0ce872254e6e      0.0s
=> => exporting config sha256:7adf35a7849c82537b81bffd9e3a9c7af4b10fa94a20a8bbd5370486bb53c506      0.0s
=> => exporting attestation manifest sha256:38737a81e455c5806c9f182412d46097022c97d02cf6391d36c0aa95fd070640 0.0s
=> => exporting manifest list sha256:f1989f771f67e88b61436ba36590ddb5a8f10780ee28cc3c900da1658dc13506 0.0s
=> => naming to docker.io/library/wireshark_lab-2_lab_environment:latest 0.0s
=> => unpacking to docker.io/library/wireshark_lab-2_lab_environment:latest 0.7s
=> resolving provenance for metadata file                                0.0s
[+] build 1/1                                                            1.1s
Image wireshark_lab-2_lab_environment Built
haris@M1-MacBook wireshark_lab % docker compose up -d
WARN[0000] /Users/haris/Developer/MSIS Manipal/SEM 1/CYT/Assignments/Lab/2_Lab/manipal_labs/labs/wireshark_lab/docker-compose.yml: the attribute 'version' is
obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 2/2
✔ Network wireshark_lab_default Created                                0.0s
✔ Container 2_lab_wireshark Started                                    0.3s
haris@M1-MacBook wireshark_lab % ifconfig -a | grep "inet "
    inet 127.0.0.1 netmask 0xffff0000
    inet 10.111.0.63 netmask 0xfffffc00 broadcast 10.111.3.255
haris@M1-MacBook wireshark_lab % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
5055459a4eac   wireshark_lab-2_lab_environment     "/usr/sbin/apache2ct..." 3 minutes ago  Up 3 minutes  0.0.0.0:80->80/tcp, [::]:80->80/tcp, 0.0.0.0:443->443
/tcp, [::]:443->443/tcp   2_lab_wireshark
haris@M1-MacBook wireshark_lab %

```

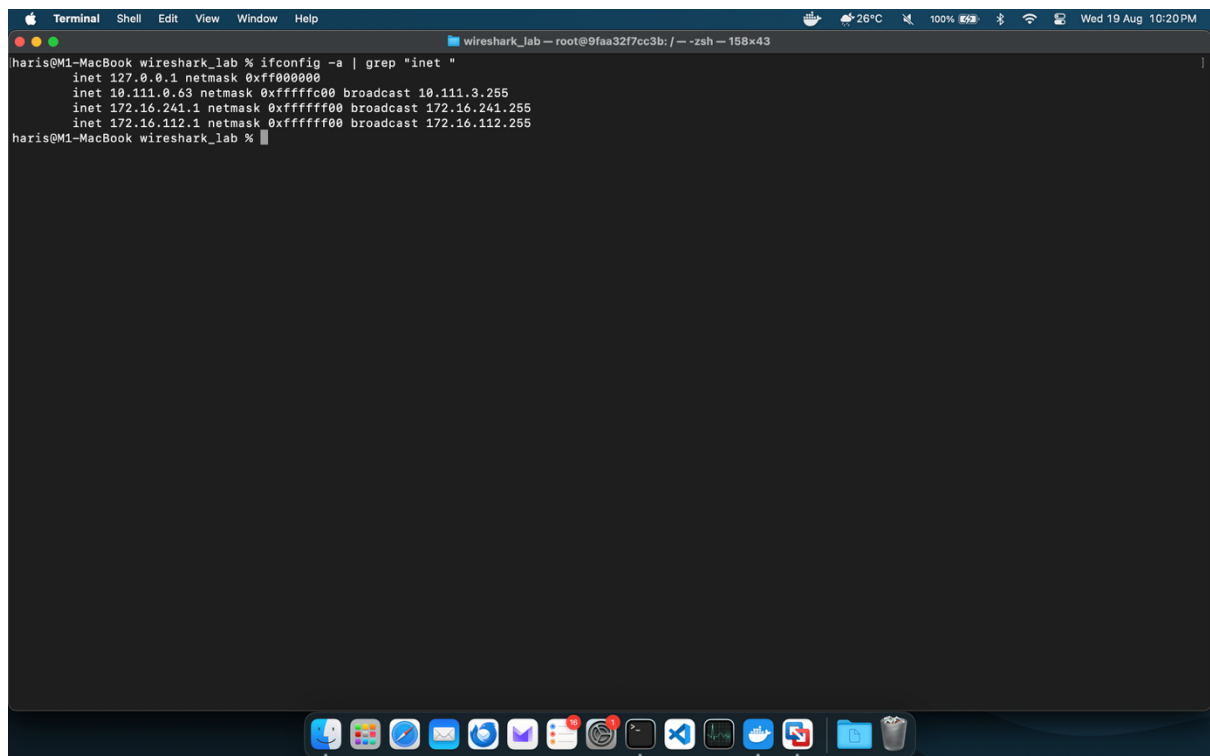
Now we have to check if the systems (attacker and victim) can communicate each other. Running *ifconfig* gives us the IP addresses of the machines

Attacker:



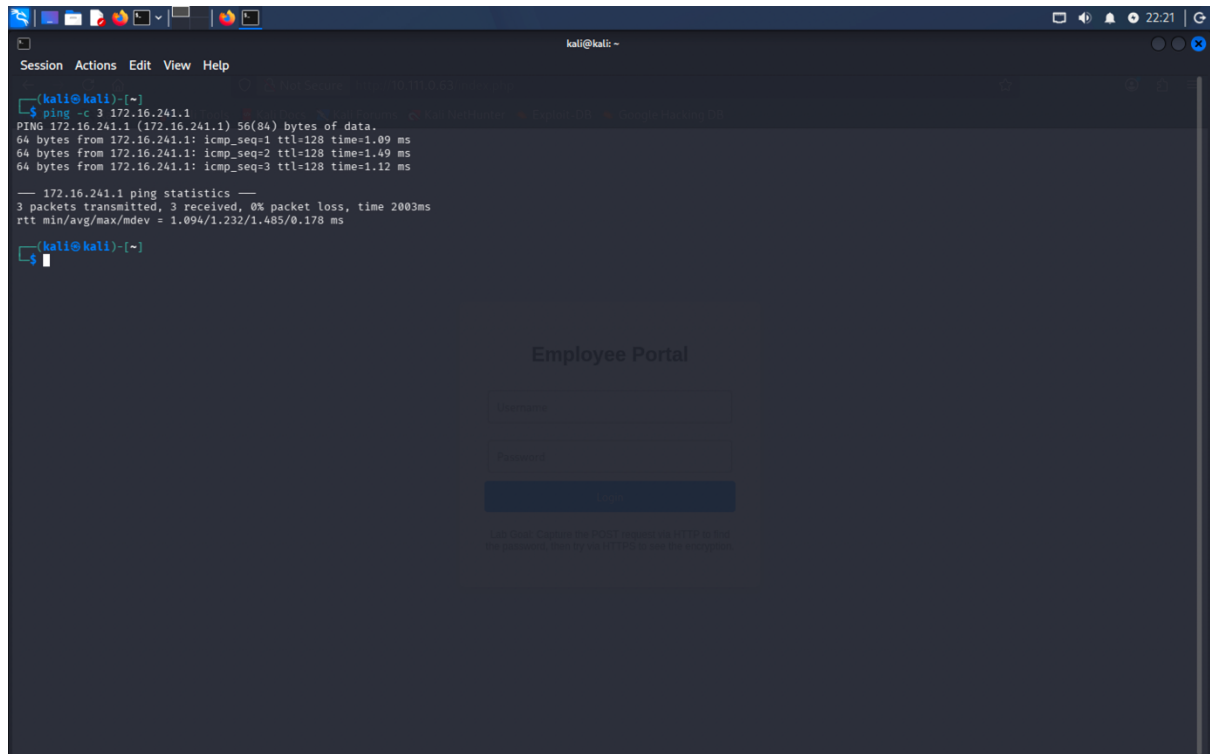
```
kali@kali: ~  
$ ifconfig  
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255  
    ether 5a:05:28:01:cd:4f txqueuelen 0 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 6 overruns 0 carrier 0 collisions 0  
  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 172.16.112.137 netmask 255.255.255.0 broadcast 172.16.112.255  
    inet6 fe80::20c:29ff:feee:d030 prefixlen 64 scopeid 0<link>  
    ether 00:0c:29:ee:d0:30 txqueuelen 1000 (Ethernet)  
    RX packets 4201 bytes 4489295 (4.2 MiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 2800 bytes 385588 (376.5 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
    device interrupt 45 memory 0x3fe00000-3fe20000  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 8 bytes 480 (480.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 8 bytes 480 (480.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Ubuntu: We are checking the IP address of Ubuntu running on docker through Mac OS.

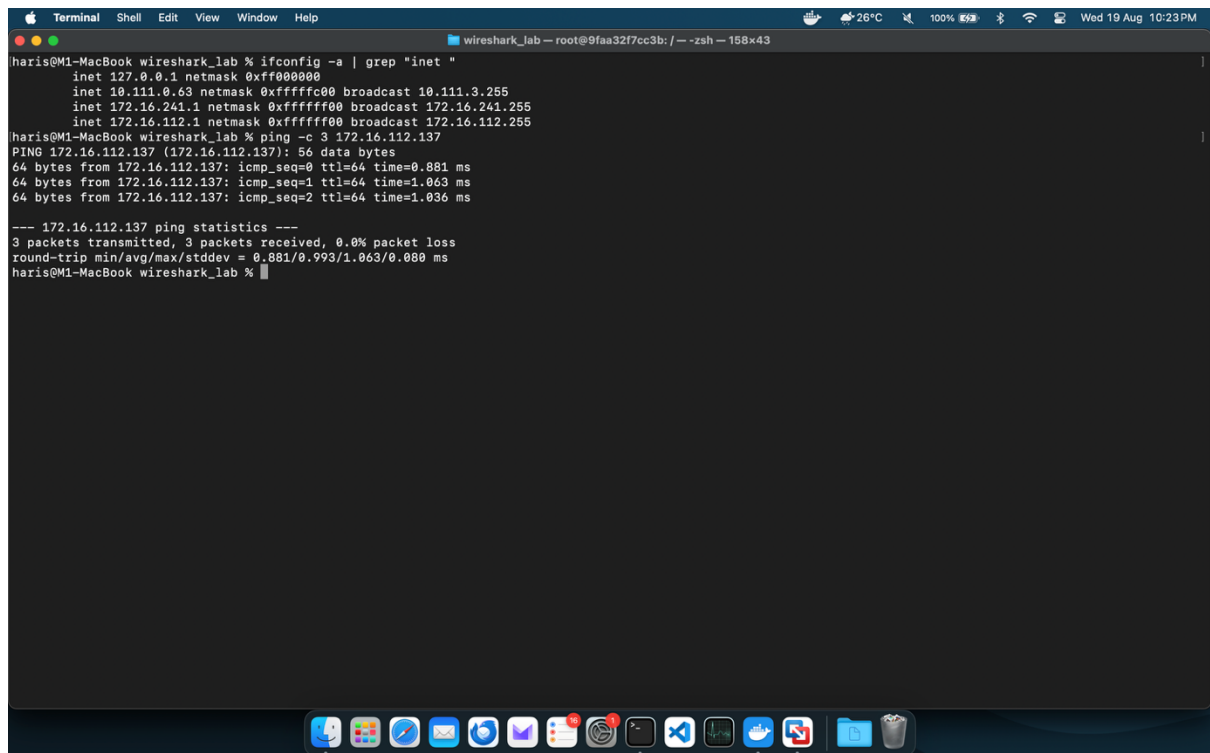


```
Terminal Shell Edit View Window Help  
wireshark_lab -- root@9faa32f7cc3b: / -- zsh -- 158x43  
harris@M1-MacBook wireshark_lab % ifconfig -a | grep "inet "  
    inet 127.0.0.1 netmask 0xff000000  
    inet 10.111.0.63 netmask 0xfffffc00 broadcast 10.111.3.255  
    inet 172.16.241.1 netmask 0xfffffff0 broadcast 172.16.241.255  
    inet 172.16.112.1 netmask 0xfffffff0 broadcast 172.16.112.255  
harris@M1-MacBook wireshark_lab %
```

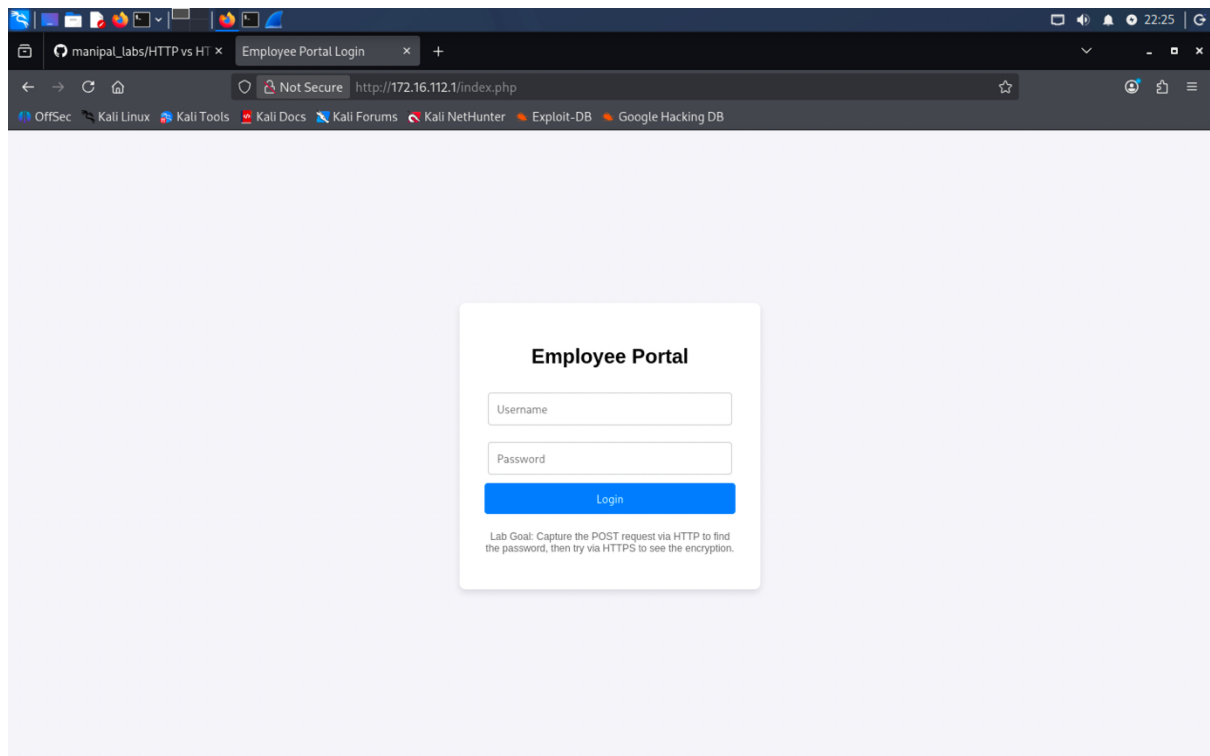
Ping from Attacker to Ubuntu:



Ping from Victim machine to Kali:

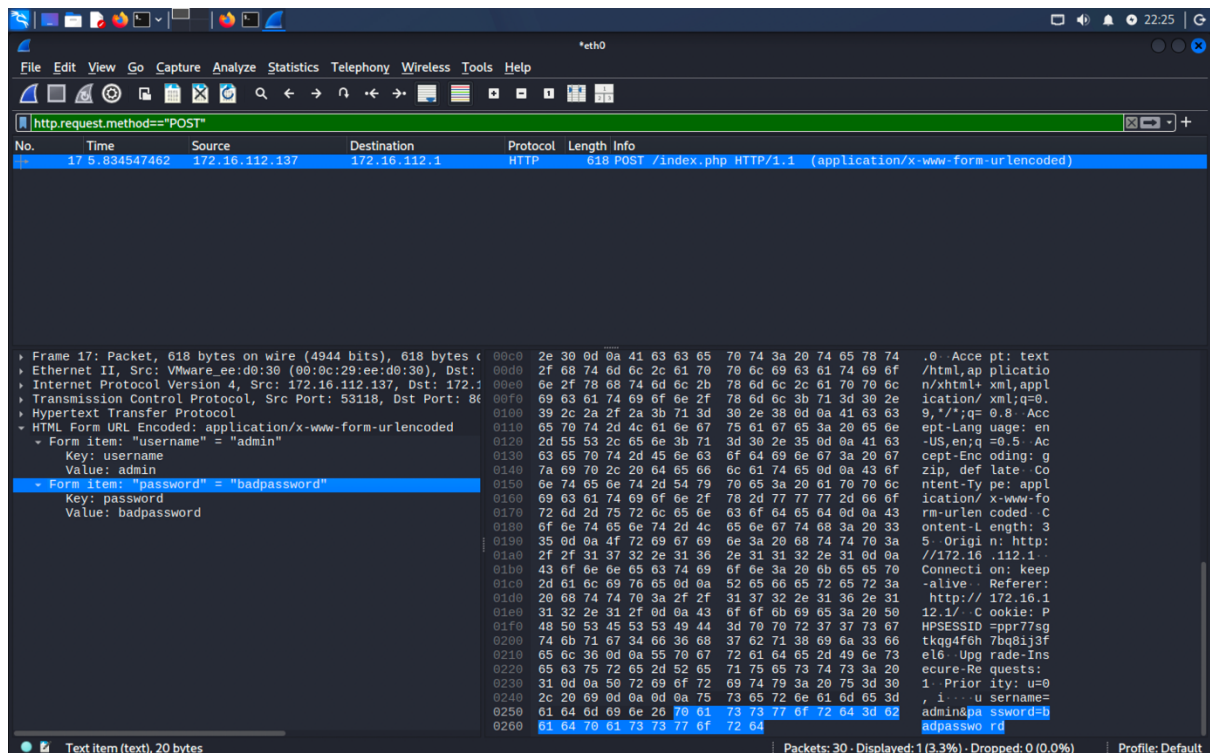


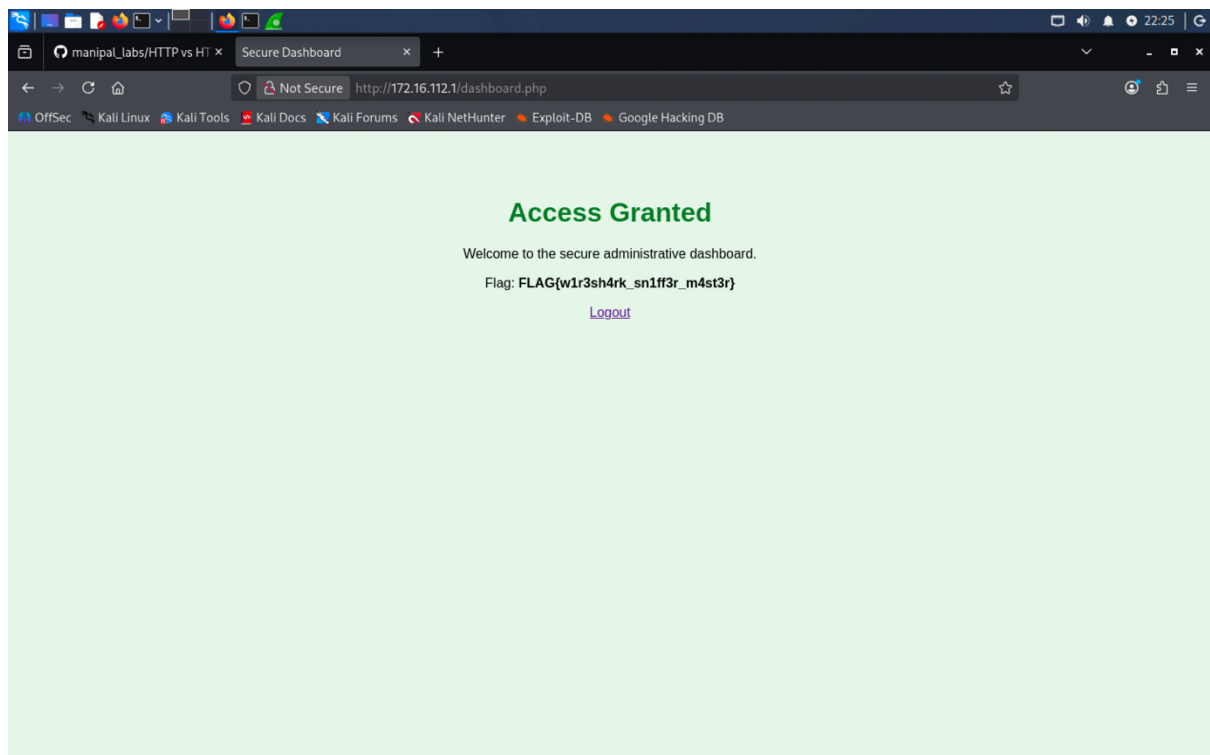
Below is the website which we are trying to access from Attacker machine hosted on Ubuntu:



Wireshark Capture of credentials on HTTP connection:

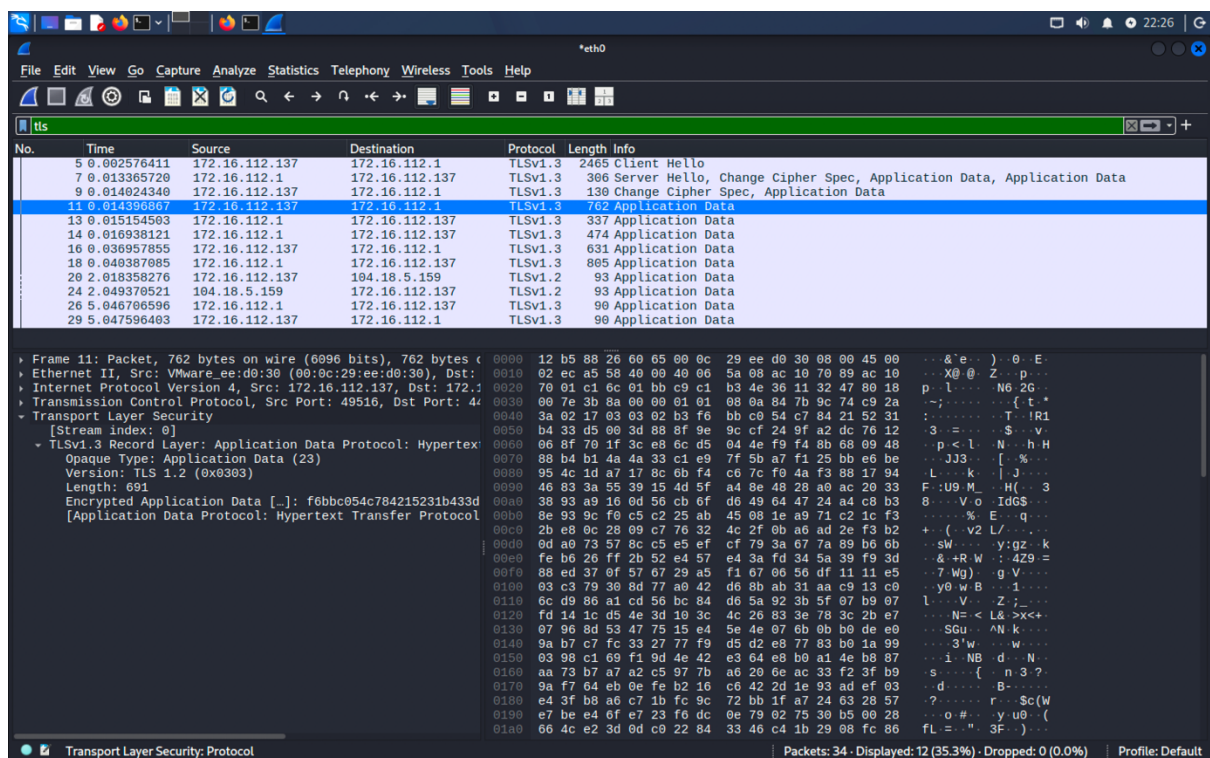
We can clearly see the credentials in plain sight under “Form Item”

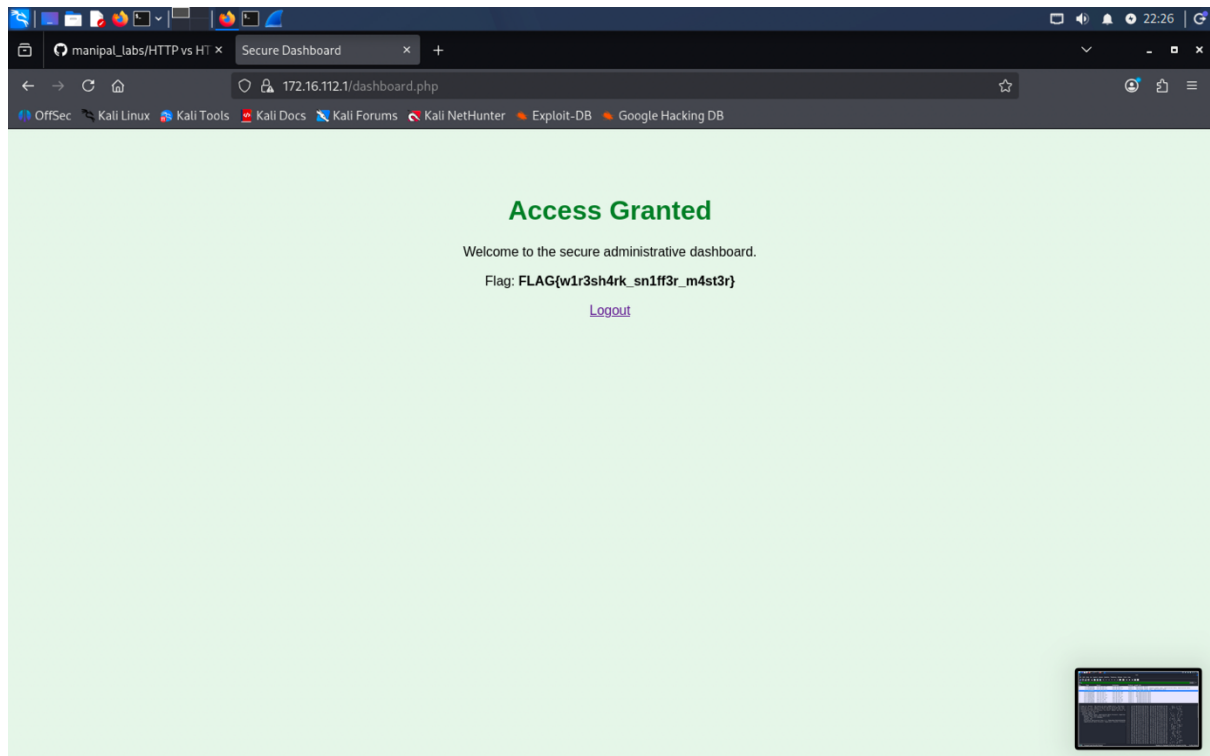




Wireshark capture of credentials on HTTPS using “tls” filter:

We can see that the credentials are encrypted and cannot be read on wireshark capture. In the figure, TLSv1.3 is being used to encrypt data in transit





We can see from the above experiment, the difference between HTTP and HTTPS traffic.

Why Encryption?

Encryption's job is to make sensitive information unreadable to anyone who intercepts it in transit. When you use HTTPS, your data is encrypted between your device and the server, so a "sniffer" on the network can capture packets but can't decipher the contents (credentials, cookies, forms, etc.). Without encryption, credentials are often sent in plaintext and can be read directly like we saw above.

It provides confidentiality in transit. It ensures intercepted traffic can't be interpreted by unauthorised parties. HTTPS achieves this by combining encryption with certificate-based server authentication, making credential sniffing ineffective for most realistic passive intercept scenarios.